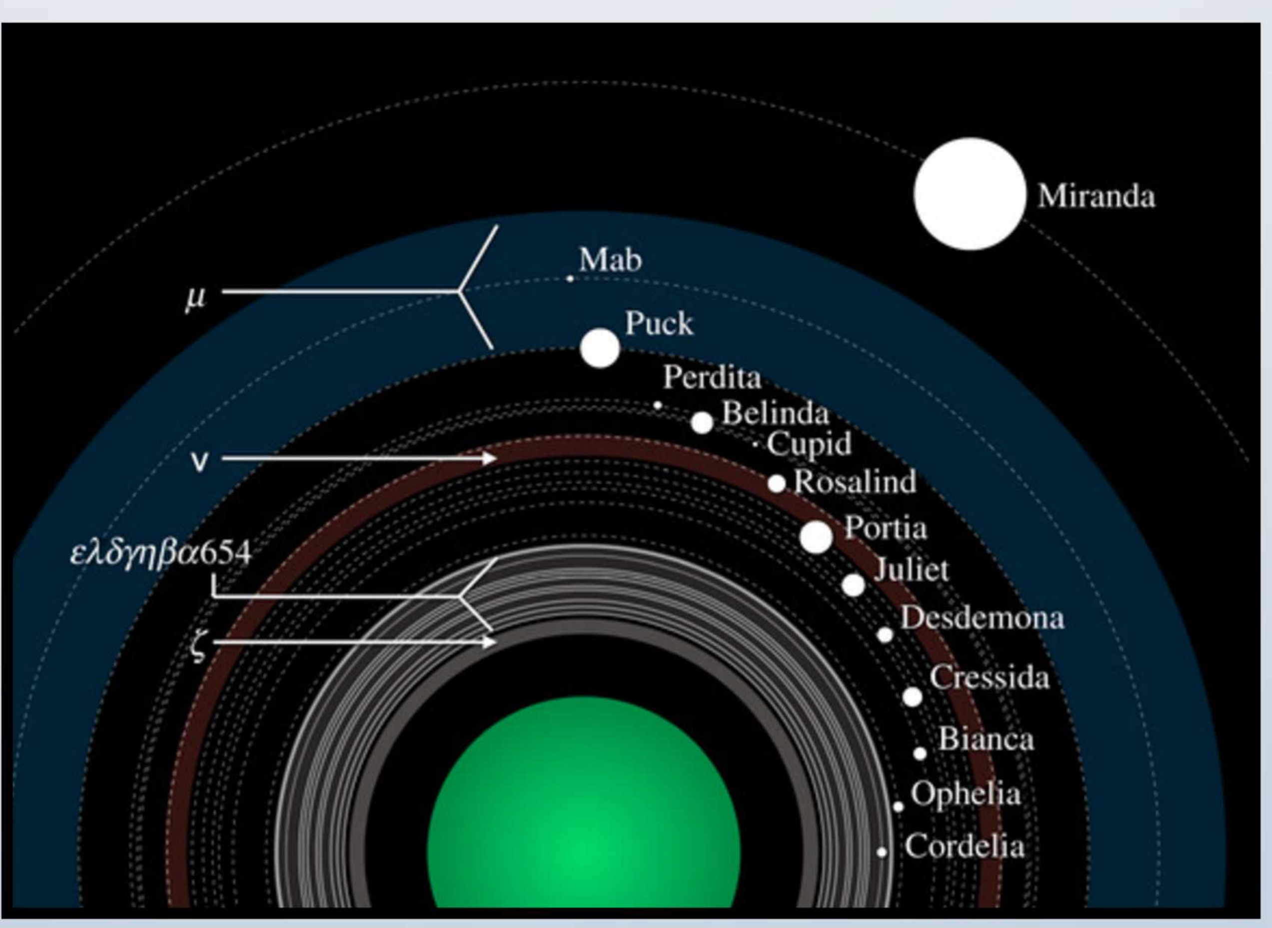
Investigating the Effects of Satellites' Eccentricity, Inclination, and Mass Ratios on their Orbits with Applications to Uranus' Moons

Introduction

- The three body problem (3BP) concerns the motion of three gravitationally interacting bodies
  - Numerical analysis is used to understand complex cases of the 3BP before analytical solutions are found
- Satellite encounter is not as studied as other cases of the 3BP (Pan et al., 2022)
  - Satellite Encounter: a subclass of the 3BP in which two small masses orbiting a large central body have a close encounter with each other
  - Three dimensional (all orbits may not be on the same plane) and highly eccentric orbits have not been studied for this problem
- The Portia group of Uranian moons is very densely packed, and, as a result, displays chaotic motion
  - Group of nine densely packed moons whose semi-major axes are separated by about 3% on average
    - Smallest difference is only about 1.2% between Belinda and Cupid
    - This configuration of moons suggests that close encounters are extremely likely to occur, and that these may affect the orbital structure



Source: Showalter (2020)

Purpose

The objective of this project is to determine the relationship between eccentricities, inclinations, and mass ratios before and after close encounters in the three body problem. Additionally, this project aims to study close encounters within the Portia group of Uranian moons.

Methodology

- Program Close Encounter Simulation
  - REBOUND
  - IAS15 ODE Integrator

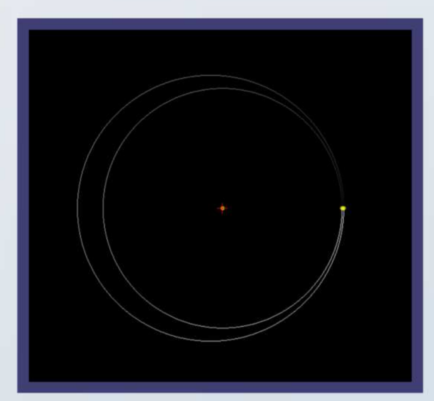
- Data Collection
  - Deterministic Sweep of Independent Variables

- Create Portia group simulation
  - MERCURIUS
    - Hybrid integrator in REBOUND

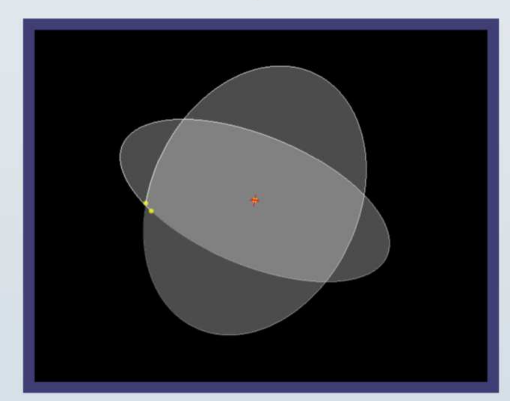
- Collect moon ephemerides from NASA JPL Horizons
  - State vectors

- Log close encounters
  - Moons, initial/final eccentricities, encounter time, minimum separation

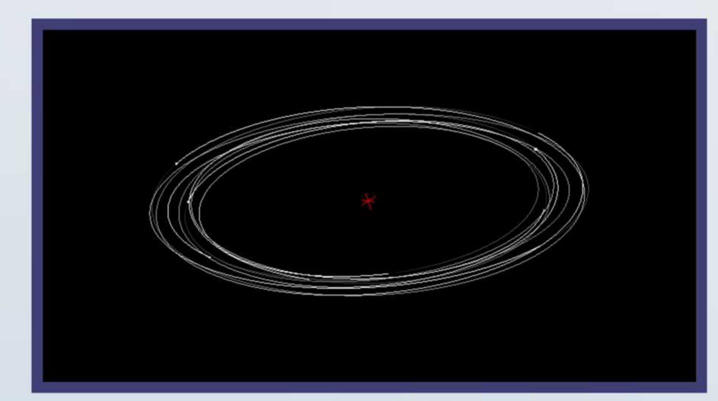
Images created by researcher using REBOUND



Varied secondary satellite eccentricity

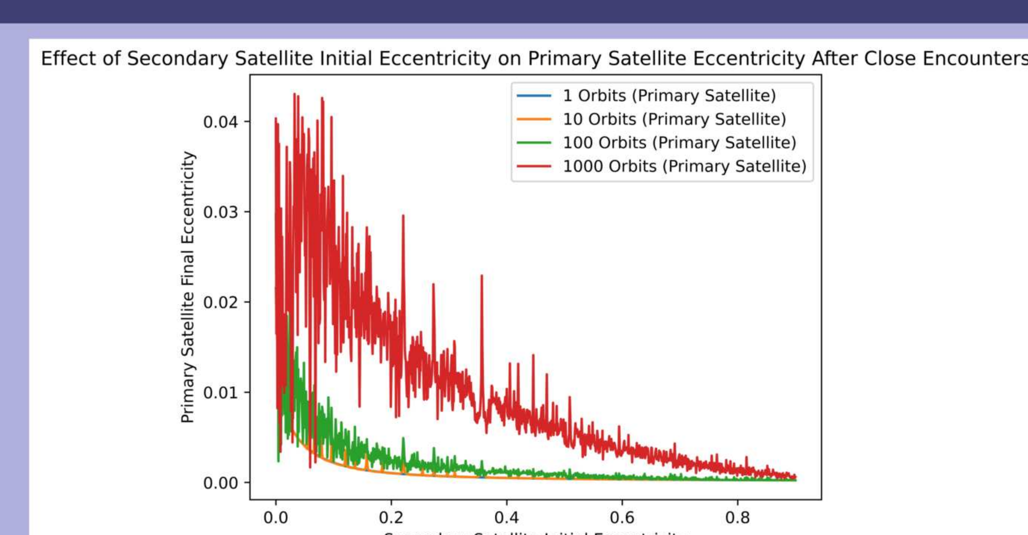


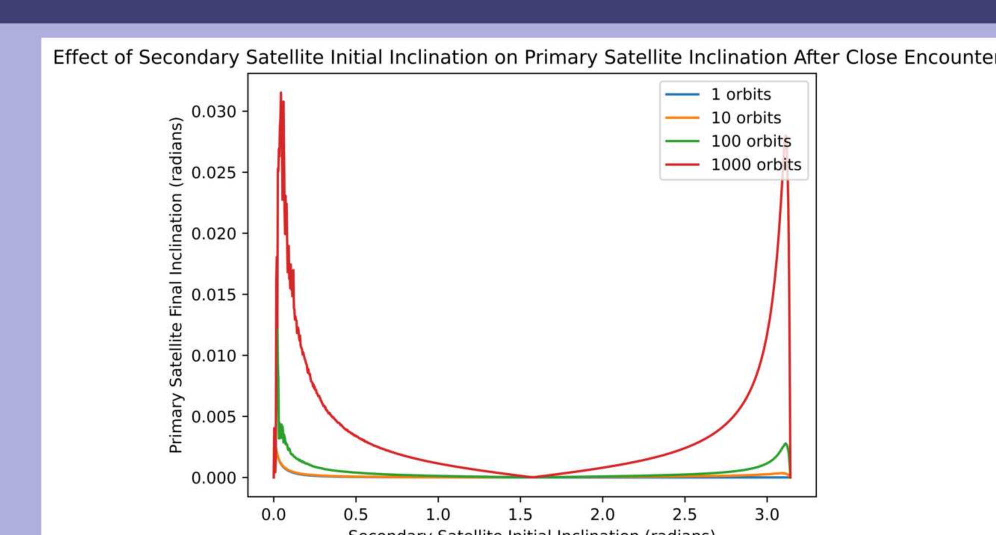
Varied secondary satellite inclination (diagonal view)

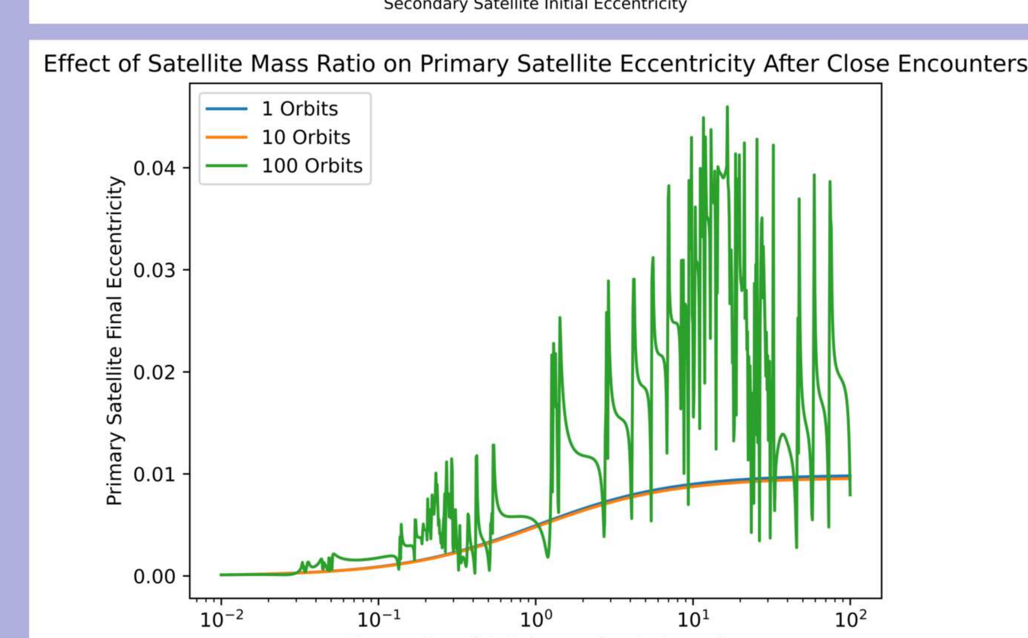


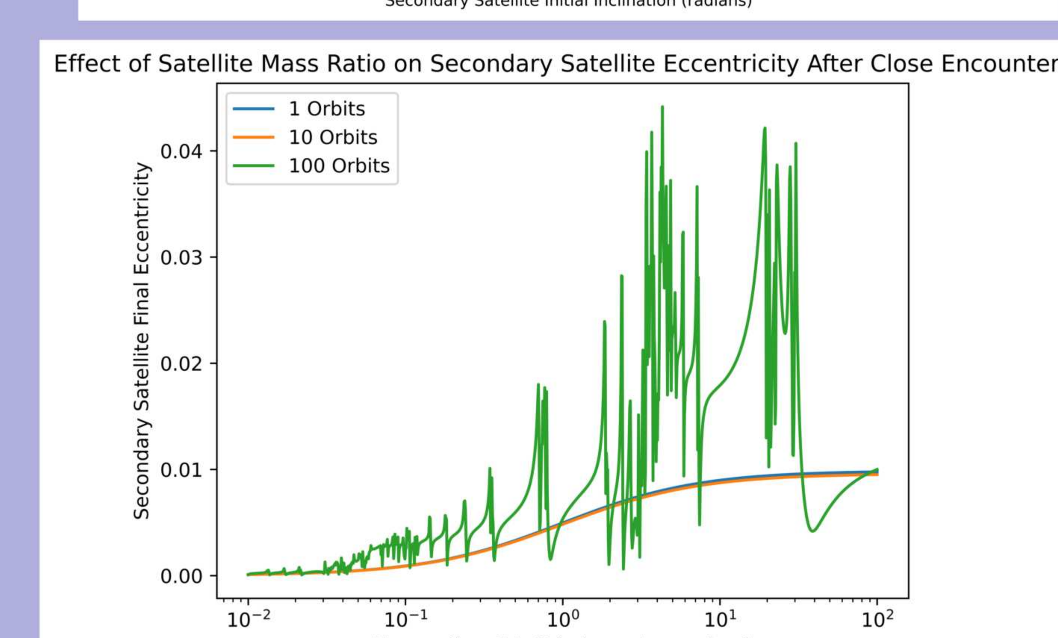
Portia group simulation (diagonal view)

Data









Graphs created by researcher using Matplotlib

Results

- Change in the primary satellite's eccentricity decreased as the secondary satellite eccentricity increased
  - Likely because the secondary satellite would spend less time near the periapsis with higher eccentricities, resulting in less total impulse delivered to the primary satellite
- As secondary initial inclinations progressed to π/2, the perturbation on the primary satellite decreased
  - Retrograde orbits seemed to show similar behavior as prograde orbits in terms of varying inclinations (though not exactly the same)
- For the Portia group simulation, on the order of 1000 years, close encounters frequently reached distances less than 0.5 of the mutual Hill's radius of the moons
  - The closest of these were encounters between Desdemona and either Juliet or Cressida

Conclusion

- Relationships between initial and final mass ratios, eccentricities, and inclinations were observed
- As systems were allowed to run for longer periods of time, chaotic motion was observed
  - Evidenced by "spiky" regions in the graphs
- Close encounters were observed and prevalent within Uranus' Portia group.

Future Work

- Analyzing chaos indicators (e.g. Lyapunov exponent) with varying eccentricities, inclinations, and mass ratios
- Creating phase portraits of Uranus' Portia group
  - Determine which close encounters lead to diverging trajectories
- Perturb initial conditions for moons in the Portia group to generate more acurate probabilistic estimates for their masses

References

- Pan, T., Xin-he, S., Xi-yun, H., & Xin-hao, L. (2022). A Review on Co-orbital Motion in Restricted and Planetary Three-body Problems. Chinese Astronomy and Astrophysics, 46(4), 346–390. https://doi.org/10.1016/j.chinastron.2022.11.008
- Showalter, M. R. (2020). The rings and small moons of Uranus and Neptune. Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 378(2187), 20190482. https://doi.org/10.1098/rsta.2019.0482